Supplementary Methods

S1 Source assets and grid record

Data-source roles and observation rules are reported in Table 1 and Sections 2.2–2.8 of the main manuscript. Exact annual Earth Engine assets are JRC/GSW1_4/YearlyHistory for 2000–2015, projects/global-surface-water/assets/GSW1_5/YearlyHistory_2016_2021 for corrected 2016–2021, and projects/global-surface-water/assets/GSW1_5/YearlyHistory_2022_2024 thereafter. Monthly assets are JRC/GSW1_4/MonthlyHistory through 2021 and projects/JRC/GSW1_5/MonthlyHistory_2022_2024 thereafter. The latter lacks system:time_start; genuine provider index/year/month metadata were used. Empty date-filter results were not dry observations.

Climate assets are UCSB-CHC/CHIRPS/V3/PENTAD and ECMWF/ERA5_LAND/MONTHLY_AGGR; radar and Dynamic World assets are COPERNICUS/S1_GRD and GOOGLE/DYNAMICWORLD/V1. The older monthly display used COPERNICUS/S2_SR_HARMONIZED. The later raw-value triage used archived Level-2A crop values and each product XML calibration, rather than reading display colors as reflectance.

The primary local raster grid preserves downloaded JRC alignment; climate intersections use EPSG:6933. The separate radar cohort grid uses EPSG:4326 with affine transform [0.00026949458523585647, 0, 0, 0, −0.00026949458523585647, 0]; its 10 m working grid is EPSG:32645. Categorical inputs use nearest-neighbor alignment and supported/water fractions use overlap-weighted aggregation. Reprojection does not add native detail or correct co-registration.

Boundary metadata are geoBoundaries gbOpen BGD ADM2, BGD-ADM2-16705992, represented year 2020, underlying Bangladesh Bureau of Statistics/OCHA ROAP, build 12 December 2023, source update 19 January 2023 and geoBoundaries commit 9469f09. The upstream license is CC BY 3.0 IGO. The original Nawabganj geometry is reported as Chapainawabganj.

S2 Sample evaluation protocol and evidence ledger

The human review component evaluated 245 complex pixel-months comprising 694 individual scene-date views across 69 Sentinel-2 scenes, recording 430 water, 236 nonwater, and 28 uncertain date-specific interpretations. Geospatial joins between human visual interpretations and sample locations rely strictly on unique sample_id and scene_id keys. The complete per-pixel validation ledger, stratum definitions, and spectral screening parameters accompany the analysis code in Supplementary Data.

The Supplementary Data tables record row counts and cryptographic SHA-256 checksums across all primary and sensitivity result datasets. Result tables, multi-sensor comparison metrics, and completed joint-cell pilot outputs are systematically cataloged in tabular manifests. Computational reproduction instructions are documented in Section S3.

S3 Reproduction

Extract Supplementary_Code_and_Tables.zip, Supplementary_Grid.zip and Supplementary_Context.zip into one empty folder, merging their Surface_Water_Study directories. Follow REVIEW_README.md for statistical, joint-cell and selected-figure replay. The supplied code, genuine inputs, sixty-nine result tables and source/checksum records support replay without Earth Engine. The monthly-2000 raster supports grid auditing, not full historical reconstruction. Upstream extraction needs authenticated access and the documented source archives; full raw histories and dated crops are not supplied. Source licenses apply.